

		Option E (incorrect). The gradient of the graph does not give any meaningful quantity.
12	A	The gravitational force on each star provides the centripetal force for the star to orbit about the common centre of mass of the system. For two stars, mass M and m, at a distance d apart, $\frac{GMm}{d^2} = m r \omega^2 = M R \omega^2$ $m r \omega^2 = M R \omega^2$ $m r = M R$ R and r are the orbital radii of the stars of masses M and m respectively. The gravitational force on each star is always directed towards the common centre of mass of the system as the stars orbit. Hence the stars should be on opposite sides of their orbital path, lying along the same straight line through the common centre between them. To maintain this, the stars must also have the same angular velocity ω. Hence star X having a larger mass should have a smaller orbital radius.

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